BIO-301 MCQs FINAL TERM- 100%

GUESS PAPER 2023

Essentials of Genetics

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OBJECTIVES

- Double mutation has a ***fitter*** phenotype than either single mutation.
- Proteins are made of "***amino acids***"
- Nucleotides contain "***ribose sugar***"
- Sequence of development "***haploid cells diploid zygote***"
- Diploid number is obtained after "***fertilization***"
- Colichine produces "***polyploidy***"
- The uses of vntr "***geomic mapping etc***"
- RNA is obtained from"***cytoplasm***"
- The type of cell division important for growth is *__mitosis__*.
- Northern blot is used for *__RNA analysis__*.
- The following is not chromosomal aberration is *__mitosis__*.
- R banding has dark bands of *__adenine and thymine content__*.
- Cancer and infertility are the result of *__genetic mutations__*.
- Colchicine is used to *__promote polyploidy__*.
- A cross for determination of the genotype of an individual is called a *__test cross*.__
- Autosomal dominant diseases are caused by *__dominant alleles on autosomal chromosomes__*.
- __Mitochondrial DNA__* is inherited through ovum, not sperm.
- Sequence difference compared to a reference standard present in at least 1-2% of a population is called *__polymorphism__*.
- Which on is sexual disease _____ __Alzheimer's disease.__
- __Discontinuous__ variations are entirely genetically controlled.
- forward genetics is defined as____ __The identification of genes responsible for a specific phenotype without prior knowledge of the gene's function.__
- genetic marker for a single locus is _____ __RFLP, Microsatellite, STS.__
- in x-linked recessive trait all are true except__ "__In X-linked recessive traits, males can be carriers of the trait."/ it is present in females more than males__
- a healthy male and female have a sòn who is color blind .what is genetics of boy parents? __XcY__

- Dark bands are those which contain ***fewer or less intense light waves***, and light bands are those which contain ***more or more intense light waves.***
- The chromosome number in an earthworm is ***2***.
- Both a husband and wife have normal vision but their son is color blind. What are the likely genotypes of the boy's parents?
- ***Boy = XaY (got recessive allele from mother) He will be color blind.***
- Position depended method of disease identification following _____***Reverse genetics***
- Molecular biology, forensic , genes mapping are application of _____***VNTR***
- Groups of human chromosomes are _____***7***
- Dna precipitation k liye konsa chemical use hota?___***Isopropanol***
- Important reagents for pcr. Option mai sb hi mojood thy tu _____***ALL***
- Mitochondrial inheritance ki cgaracteristics di thi option mai. Ghalat konsi ye btana tha. These are present chose other than these ***(● Heteroplasy ● Variable expression ● Pleiotropy ● Reduced penetrance ●Functional somatic mosaic***).
- ***Q-PCR*** is commonly used to determine whether a DNA sequence is present in a sample and the number of its copies in the sample.
- Meiosis I ki characteristics pochi thi option mai. ***Ye b ALL***
- In which pcr two pairs of primers are used._____***Nested PCR***
- In giemsa staining dark bands are rich in... ***A and T ans***
- And light bands are rich in***.... G and C ans***
- During rna extraction, rna is extracted from the neucli of which cell. Rbc, ***wbc***, or platelets
- ***Erythrocytes*** k bry mak 1 mcqs tha_____ Erythrocytes (red blood cells) do not contain nuclei. They neither synthesize nor contain RNA.
- Which fur color of rabbit have highest numbers of genotypes. ***Grey***
- Which genetic disorder have lowest frequency. ***Ans autosomal recessive disorder***
- Which is an autosomal recessive disorder? ***(● Bloom syndrome ● Carpenter syndrome ● Cystic fibrosis ● Thalasemia ●Many forms of mental retardation***)
- Which mutation shortens the protein? ***Ans non-sense mutation***
- How many cases of turner syndrome are due to ring chromosome. Ans ***20%***
- During inversion what is reversed.___***chromosome sequence/segments of chromosomes:***
- Which mutation suppreses the effect of other mutation. Ans ***suppresion mutation***
- ***DOMINANT EPISTASIS:*** Epistatic gene exerts its affect with the presence of a dominant allele.
- Mutation that occur in regulatory region affects? ***Ans transcription, Translation and splicing of protein***
- ***AUTOSOMAL DOMINANT DISEASES*** (● Marfan syndrome ● Huntington's disease ● Retinoblastoma ●Waardenburg syndrome ●Myotonic dystrophy ● Polycystic kidney disease ● Achondroplasia ● Polydactyly ● Heredity hearing loss)
- ***X-LINKED DOMINANT DISEASES:*** (● Rett syndrome ● Alport syndrome ● Goltz syndrome ● Fragile-X Syndrome)
- Homologous paired chromosomes have the same shape, size, and location of the ***centromere***.
- ✨RNA is present in....(***ans) Few viruses***
- in organic method of purification of DNA ..... IS USED TO ***remove protein (ans) phenol chloroform***

- ..... Is chelating agent(ans)***EDTA***
- ..... Result in two cells(ans) ***mitosis***
- application thi VNTR ki uska answer bh ***VNTR hi th***
- Dark region rich in .... (***At***)and light in..... (***GC***)
- R banding has dark bands of: (ans) ***Heterochromatic***
- Fur color in labrador retriever controlled by….(ans)***2gene***
- Two pairs of primer use in…(ans) ***nested PCR***
- Male pattern baldness is a recessive sex-linked trait on the X chromosome. A woman, whose father had male pattern baldness, marries a man with this trait. What is the probability that any son born will have pattern baldness?.. (***ans)50%***
- A mutation that are responsible for shorten of protein due to producing stop codons***….(ans)non sense mutation***
- protein that is used to packed DNA?... (ans) ***Histones***
- when allele of two gene segregate together and transmitted as a unit? What type of segregation occur... (Ans) ***Dependent segregation***
- Chance of inheriting a mutant allele from one parent is 50% and for the other parent is also 50%. Net chance of inheritance of two mutant alleles is ***....(ans)25%.***
- dna fingerprinting is an example of??***VnTR***
- If normal male and normal female are not color blind but their son is color blind what are the alleles of next generation: ***A)*** **Son carry recessive allele**
- Hemophilia is an example of___***X linked chromosome.***
- Two pairs of primer use in….. ***nested pcr***
- Genes in drosophila.....***13,700***
- A mutation that are responsible for shorten of protein..... ***Nonsense mutation***
- ..... Caused frameshift mutation._____ ***Deletion***
- First stage of mitosis.....***Prophase***
- First phase of mitosis.....***Prophase***
- ..... is non-ionic buffer for DNA extraction (***cell lysis buffer***)
- If both parents have heterozygous dominant disease, what will be the chance of a child not having that disease? (***25%***)
- microbiology, finger printing, genetic diversity and mapping of genomes are applications of.....***VNTR***.
- Symbol of substitution ____ ***">"***
- RNA is genetic material in.....***some viruses***
- A type of genetic study in which , select a gene create a mutation then study its phenotype ***..... Forward genetics.***
- If two people who are both carrier for the genetically inherited fatal recessive disease decide to become parents , how many chances that their children will also be carriers.(***1 out of 4***)
- Any change in nucleotide sequence of DNA gene is called.....***mutation***
- Which disorder have the lowest frequency range.....***X linked disorder.***
- In phenol- chloroform DNA extraction lower organic phase contains.....***lipids***
- DNA finger printing bioinformatics are applications of (ans)***VNTR***
- Dark region rich in .... (***At***)and light in..... (***GC***)
- R banding has dark bands of: (ans) ***Heterochromatic***
- Two pairs of primer use in…(ans) ***nested PCR***
- Male pattern baldness is a recessive sex-linked trait on the X chromosome. A woman, whose father had male pattern baldness, marries a man with this trait. What is the probability that any son born will have pattern baldness?50%
- A mutation that are responsible for shorten of protein due to producing stop codons….(ans)***non sense mutation***

- protein that is used to packed DNA?... (ans) ***Histones***
- Chance of inheriting a mutant allele from one parent is 50% and for the other parent is also 50%. Net chance of inheritance of two mutant alleles is ....(ans)***25***%.
- Methods for DNA quantificationAns: ***organic and inorganic both***
- Blood sample for RNA purification can taken from: Ans: ***RBCs / plasma or serum***
- Light and dark color of chromosomes is trah aik line h Ans: ***light color or bright color contains adenine and thymine while dark color contains cytosine and guanine.***
- Applications of vntr....***ALL***
- RNA is genetic material in Ans: ***Only few viruses***
- pidegree sign:⊘ Ans: ***death***
- nucleosome is composed of a core of ***eight*** histone proteins and DNA (***147 bp***) wrapped around them.
- Point mutation for ***single base pair***
- Flemming discovered ***Lambrush***
- Lampbrush was discovered in *****1882***
- f the (1 base) insertion the swift correct statement is " ***THE DOO GRA NOU***"
- RNA is present in....(ans) ***Few viruses***
- Dark region rich in .... (***At***)and light in..... (***GC***)
- banding has dark bands of: (ans) ***Heterochromatic***
- one parent is 50% and for the other parent is also 50%. Net chance of inheritance of two mutant alleles is .... (ans)***25 %.***
- Tetrads align at center… ***Metaphase I***
- Frequency of autosomal recessive disorders per ***1000*** individuals is. ***2-2.5***
- Microbiology DNA fingerprinting Genetic diversity Forensics and Mapping of genomes are applications of ***VNTR***
- PT cross with Tp…will result ***TP 50% To 50%***
- During RNA purification, RNA pellet is dissolved by adding…..to each sample. ***RNAse-free H20***
- Both a husband and wife have normal vision but their son is color blind.What are the likely genotypes of the boy's parents?***Boy recessive****
- of DNA purification…. (***Organic+inorganic) Both***
- is commonly used to determine whether a DNA sequence is present in a sample and number of its copies in the sample. ***Q-PCR***
- In standard PCR procedure the required annealing temperature is In mutation a single base pair is altered….. ***40-68°C***
- Major ingredients of Polymerase Chain Reaction (PCR) is/are:…… ***buffer,mgcl2, primers, DNA template***
- Is a X linked fragile (***46XY fra X , q27.3***)
- Sugar in DNA is ***Pentose***
- Chromosome are the one having centromere and ***2 arms***
- Don't have RNA. ***Erythrocytes***
- Phase between G1 and G2 *****S phase***
- which transfer mutation ***Germline***
- nutrition is given to fetus by ***Mother***
- results from suppressor mutations that occur in the different gene as the original mutation. Intergenic. ***suppression***
- Addition mutation is actually results in a ***frame shift mutation***
- When the entire genomic sequence is not known, a cDNA reference sequence should be used. ***First***.

- ⊘ Is symbol of ***Death*** ☠
- The lysis buffer destabilize the? ***Cell membranes***
- Which is not correct female effect more
- Mutation at lower case, ***RNA level***
- ***EDTA***..... Is a chelating agent.
- The purpose of southern blot test is to .....***DNA***
- In what which type of polyploidy is found .... ***autopolyploidy***
- Splitting of sister chromatids occur at ..... ***anaphase***
- Paired chromosome with same length shape and centromere location .....***Aulusome***
- Giant chromosome of Drosophilla have ***1024*** strands. In case of Chironomous ***4096***.
- 30% genes diallelic exist in form-----***2 Allele A and B Codominance***
- Mt DNA genes in human ***37***
- Deletions of larger proportion are usually incompatible _____***with life***
- Every individual have ***6*** mutation each cell
- Purine is replaced by pyrimide_____***Tonsversion*** then
- ***Addition*** __mutation results in frameshift
- RNA is present in ***few viruses***.
- "_" (underscore) indicates a range of affected residues, separating the first and last residue affected Amniotic fluid analysis to determine genetic and chromosomal disorders after ***14 weeks gestation***.
- ***Polymorphism*** _____can be clue to Single of thousands of bases
- Amplified fragment run in agarose gel detected by ***Ethidium bromide***
- Annealing temp. ***40_68°C/40-65***
- methods for RNA purification **A organic (b) inorganic (c) both**
- A woman is heterozygous for Tay-Sachs (a recessive disorder). What are the chances of her giving birth to a child with Tay-Sachs; if the father is normal (homozygous dominant?) What are the chances of giving birth to a carrier?
- **Woman: Tt, Man: TT, No chance of child with disease 50% chance of carrier.**
- Both a husband and wife have normal vision but their son is color blind. What are the likely genotypes of the boy's parents?
- **Boy = XaY (recessive from mother = colorblind son)**
- A man with blood group A, marries a woman with blood group B. Their child has group O blood. What are the genotypes of parents? What other genotypes and in what frequencies would you expect in offspring from this marriage?
- **Father must be I BI i and the mother must be I AI i or they would be vice versa.**

- who discovered genetics .. ***Gregor Mendel***
- Two Barr bodies are due to deactivation of two____ ***X chromosomes***
- Which is not Autosomal dominant disease? __***Gaucher's disease***
- In ZW-ZZ system, which is male? ___***ZZ is male***
- Down syndrome is due to trisomy of ___***chromosome 21***.
- _____is non-ionic detergent, designed to lyse outer cell membrane of Blood and epithelial cells, but will not break down nuclear membrane. ___***Cell Lysis Buffer***
- Darker bands in R staining are ___***Euchromatin.***
- Annealing temperature in PCR ___***40-65°C.***
- ____ is commonly used to determine whether a DNA sequence is present in a sample and number of its copies in the sample___ ***Q-PCR***
- ___used for identifying of SNPs.__ ***allele specific PCR***
- Whole genomes can be amplified by ___***Whole genome PCR***.

- Mini Primer PCR can extend from ____***short primers.***
- Northern blot is a method used for detection of a specific ____sequence.__ ***RNA***
- Which describes best the prokaryotic chromosome ____ ***Single, circular DNA molecule.***
- Prokaryotic nucleus lies in ___***nucleoid.***____are used to determine the pattern of inheritance of a genetic disease within a family.__ ***Pedigrees***
- Variations are due to ___***Genes & environment.***
- In ring chromosome, two ___ joined joins to form ring shape. ___ ***arms***
- Human zygote is ___ chromosomes. ***Diploid and 46***
- ___ Histone tightly pack the nucleosome.__***H1***
- Z-DNA is ____ helix. __***Left-handed***
- Which is Transition mutation?___ Purine is replaced by purine.Or Pyrimidine is replaced by pyrimidine.____ ***Both are correct.***
- Growth takes place by ____***Mitosis***.
- ____push against each other, that stretch the cell into an oval shape in Anaphase. ___ ***Non-kinetochore spindle fibers***
- Meiosis I results in two ___ daughter cells. __ ***haploid***
- Cystic fibrosis and muscular dystrophy are ___***Single gene disorders.***
- Genes switched on when have ___ cytosines.__***unmethylated***
- Trisomy is ___ ***Aneuploidy***.
- ____produces a change in amino acid sequence in protein product.___ ***Missense*** mutation
- Which is not true about Autosomal recessive disease?____ Diseases occurs in individuals with two mutant alleles. ***This one is not true***.
- Homozygotes with two copies of the altered gene. ***It's true.***
- Female bee sex is_____***haploid***
- Recovery of DNA is called_____ ***DNA extraction or DNA isolation***
- Autosomal recessive _____***cystic fibrosis***
- Mendel's work_____ ***3 laws***
- X0 system in_____***in***sects
- Xo trait s present in_____***fruitfly***
- Nature has variations ____ ***widespread***
- "m." For a mitochondrial sequence _____(***like m.8993T>C,)***
- Rabbit having light grey colour_____***chinchila***
- Polygenic traits are----(***qualitative***)
- In Giemsa staining Dark region bands are----- and light region bands are***----- (A-T and G-C).***
- Arm of the chromosomes are named as------ ***(p and q)***
- Genetically determined variability among humans is due to a difference in---- of the genomic ***sequence. (0.1%)***
- In TELOPHASE I------set of chromosomes arrived at each of the pole***.____(Haploid)***
- …..is used for identifying of SNPs__ ***Genotyping***
- Annealing temperature….. ***40-65°C***
- in R banding darker region of chromosome represent…… ***Euchromatin***
- origin of …….type of mutation unknown__ ***spontaneous mutation***
- cri- du- chat syndrom is caused by …***Mutation***.
- monosomy represent as ….. base___ ***2n-1***
- lampbrush chromosome are found in… ***1882***
- Fragile X syndrome is …. ***X-linked dominant disease***
- Epistasis is an interaction of how many genes.. ***two or more***

- HGNC stands for… (***HUGO gene Nomenclature committee***)
- Frequency of ___ is highest. ___ ***autosomal dominant disorders***
- Microbiology, DNA fingerprinting, Genetic diversity, Forensics, Mapping of genomes are applications of?___***VNTRs***.
- Male pattern baldness is a recessive sex-linked trait on the X chromosome. A woman, whose father had male pattern baldness, marries a man with this trait. What is the probability that any son born will have pattern baldness?
- ***The probability that a son will be born with pattern baldness is 50%.***
- Chromosome number in round worm__ ***2***
- In which stage of Meiosis I chormosomes are replicated..? __***Anaphase***
- In which phase of meiosis, chormosomes are replicated. (***Options meiosis I, meiosis II, Both***
- Chromosome can be counted only in ….. Phase..? __***Metaphase***
- ___Can change color of hydrogenae flower.__ ***PH of soil***
- Which part of chromosome is visible when viewed under microscope… ***chromatid***
- ___cases of turner syndrome are due to ring chromosome.___ ***20%***
- Annealing temperature in PCR ..***40- 65°C***
- Which blood group is universal recipient___ ***AB blood group***
- About 25000 SNA in human are _____ ***non- synonymous***
- A condition in which some people with an appropriate genotype fail to express the phenotupe is called as _____***penetrance***
- Pattern baldness is a recessive trait. A female inherits it from her father and marries a man who has this trait . What are the chances to pass on this trait
- ***50%***
- Expression of phenotype by a number of individuals is called as_____ ***phenotypic*** variation
- Geimsa stain is used for ___ chromosomal banding ***(C-banding)***
- Northern blotting is used for____ ***RNA***
- Southern blot is use for ____ ***DNA***
- Cri- du- chat syndrom is caused by … ***mutation***.
- The Type of cell division important for growth is___ ***mitosis***
- Both a husband and wife have normal vision but their son is color blind. What are the likely genotypes of the boy's parents? ***Boy = XaY (recessive from mother = Colorblind son)***
- Blood cells are lysed ___***erythrocytes lysis and leukocytes lysis.***
- Dissolve RNA pellet by adding ___***RNAse-free H2O or buffer***
- In PCR, DNA is ___ ***amplified***.
- Which is an exception in Mendelian inheritance ___***multiple alleles***
- Trisomy 18 causes which disease ___***Edward syndrome***
- Who discovered gene map in Drosophilla?__***T.H Morgan***
- How many no.of genes are present in human mitochondrial genome __***37***.
- Which of the following is not an example of multi-locus marker ___SST
- Which PCR is used to determine SNPs ___ ***Allelic specific PCR***
- Dark bands are rich in __***A.T*** light bands are rich in ___***G.C***
- Whole genomic size & amplification is done by ___***Whole genome PCR***
- Fur color in Labrador retriever controlled by ___***2 genes.***
- Annealing temp__ ***40-65°C***
- ___ is used to identify region by using probes.__***Southern blotting***
- Northern blotting is used to for ___ ***RNA***
- Okazaki fragments occur in ___***discontinuous replication***
- The cross of PPTT and pptt, F1 dominant progeny percentage is___***100%.***

- ⊘ Symbol means___ ***dead***
- Which is not related to X-linked trait…? ***it is present in females more than male***
- R banding has dark bands of____ euchromatic ***(G-C)***
- cancer, infertility are result of _____ ***Multifactorial disease***
- Colchicine has the extremely useful property of preventing formation of the ___***Spindle,***
- Autosomal dominant diseases____ ***Huntington's disease***
- ____is inherited through ovum not sperm__ ***Mitochondrial DNA***
- Standard present in at least 1–2% of a population_____ ***DNA polymorphism***
- A smooth curve is property of …***Continuous variation***
- DNA replication is …. ***Semi conservative***
- In case of X linked recessive disorder A Cross b/w abnormal and normal female Will be … ***50%***
- Multifactorial disease caused by….. ***genes and environmental effect***
- Addition mutation is the addition of….. ***nucleotides/base pairs***
- ……is a chelating agent.__ ***EDTA***
- PCR is a technique used for nucleic acid… ***Amplification***
- Most common inversion seen in human is on chromosomes……..***at inv p12 q13 __ nine***
- The disease is said to have ….If all individual who have disease caused mutation have clinical symptoms . ***Complete penetrance***
- For high resolution chromosomes analysis which stages of cell cycle are cell arrested before staining….. ***metaphase stage***
- How many cases of Turner syndrome are due to using chromosomes ***… 20%***
- Which type of mutation occurs boundaries of exons and intron … ***splice mutation***
- Which disease is due to deletion in chromosomes ***… Cri-du-chat syndrome***
- An example of autosomal dominant disorder is …..***Huntington's disease***
- During DNA purification RNA pellets is dissolved by adding……to each sample__ ***RNase free water or buffer***
- In organic method of DNA purification….is used to remove the protein.__ ***phenol chloroform***
- Whole genome application PCR can be used to amplify … ***whole Genome***
- …..is used for identify SNPs .__ ***genotyping***
- Method for DNA purification include … (***organic, inorganic, chelex, FTA, silica based***)
- Blood cells are type in …. ***Blood typing Procedure*** .
- Prokaryotic nucleus lies in __***nucleoid***.
- Unit of genetic map……. ***Centimorgan***
- Who discover Restrictions Endonuclease…… ***Arber, Smith&Nathans***
- Protein activate after dna destroy……. ***Histone***
- DNA Replicate at which stage of interphase……. ***S phase***
- Division of cytoplasm called…… ***Cytokinesis***
- An average cell can divide ……. Before it Gets unhealthy__. ***50 Times***
- The unspecialized cell that renew …… ***Stem Cells***
- Bones regenerate every…… ***10 years***
- Golgi apparatus found in…… ***Eukaryotic Cells***
- JAPANESE 4 O' CLOCK FLOWERS example of___ ***incomplete dominance***
- Strawberry is___ ***octaploid***
- Smooth curve___ ***continuous variations***

- In individuals with the SRY gene, the generic embryonic gonads are modified into ___***testes***.
- Chromosomes……lead to down syndrome. Chromosome 21 113. Monosomy represent as ……. ***Base . 2n-1.***
- If 14 % nucleotide from DNA molecule contains base T what present will contain base G ??.... ***36%***
- Cell splits into two in …… ***mitosis***
- Types of cell division important for growth is … ***cytokinesis or telophase***
- Haploid gametes are formed by ….. ***meiosis***
- How many numbers of alleles control of color of rabbit …… ***four***
- It is estimate that about ……Genes are imprinted can found on several different chromosomes. ***40***
- CML stand for …. ***Myelogenous leukemia***
- Which type of mutation involved in significantly reduce the longivty of individual…. ***Lethal mutation***
- Which is example of X linked disorder…. ***Hunter's disease***
- In pedegre analysis symbol ⊘represent…… ***dead***
- Which is not example of multi locus markes…. ***STS***
- Mutation can be identify by ……***blotting***
- Chromosomes have centromere at one end of the arms … ***telocentric***
- Ants and bees have …. ***No sex chromosome***
- About 25000 SNPs in humans are … ***non- synonymous***
- Lamp brush chromosomes were discovered in …. ***1882***
- The idea that different pairs of alleles are passed to offspring independently Mendel principal of …. ***Law of independent assortment***
- Mendelian inheritance the gene that obey ….. ***principal o segregation and independent assortment***
- Skin color of pigment in humans is controlled by at least …. Separately inherited genes . ***Three***
- ……banding pattern of chromosomes used variety of technique at stain tha ***GC***
- Rich DNA on the chromosomes.__***Giemsa staining***
- The nucleosome is composed of a score of ……. ***Histone proteins and DNA***
- ……BP wrapped around them. ___***8, and 147***
- How many types of histone molecules in nature ?……. ***Eight***
- Which can be seen using the microscope during karyotyping……. Chromosom***es***
- More than two sets of chromosomes which are coming from different parents…….***Polyploidy***
- Monosomy represent as…. ***2n-1***
- During…..of DNA the coiling must be released.__ ***transcription***
- During recessive epistasis …gene exerts its effect with…. Genotype .__ ***one, homozygous recessive***
- ■—● Symbol means___ ***mating***
- Which one of the statement is correct about colour blindness is___***Affected individual can't differentiate the Green and red colour***
- Female bee sex is___***diploid***
- Annealing temperature….. ***40-68°C***
- Mini Primer PCR can extend from __***short primers.***
- Darker bands in R staining are ___***Euchromatin.***
- Male pattern baldness is a recessive sex-linked trait on the X chromosome. A woman, whose father had male pattern baldness, marries a man with this trait. What is the

probability that any son born will have pattern baldness? ***The probability that a son will be born with pattern baldness is_____50%.***

- Both a husband and wife have normal vision but their son is color blind. What are the likely genotypes of the boy's parents? ___ ***recessive from mother***
- Major types of mutations___ ***3***
- Southern blotting____ ***Identified DNA may be a small piece of DNA or a mutation.***
- __ relieves the effects of a mutation in one gene by a mutation somewhere else within the genome. __***Intergenic suppression***
- Microbiology, DNA fingerprinting, Genetic diversity, Forensics, Mapping of genomes are applications of?___***VNTRs***.